AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 8 · STUDENT LESSON

Structured Output and Schemas

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Before you start

AI-2 Week 7: text and numeric values are different types.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Separate JSON parsing, schema validation and source support checks. Try the recall before opening notes.

Your learning target

I can separate JSON parsing, **schema validation** and source support checks.

I can reject invalid **type**s and unexpected **field**s under a **closed schema**.

I can trace a validly formatted but unsupported claim.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Iterative Prompting. Give an example and a mistake that the idea helps you avoid.

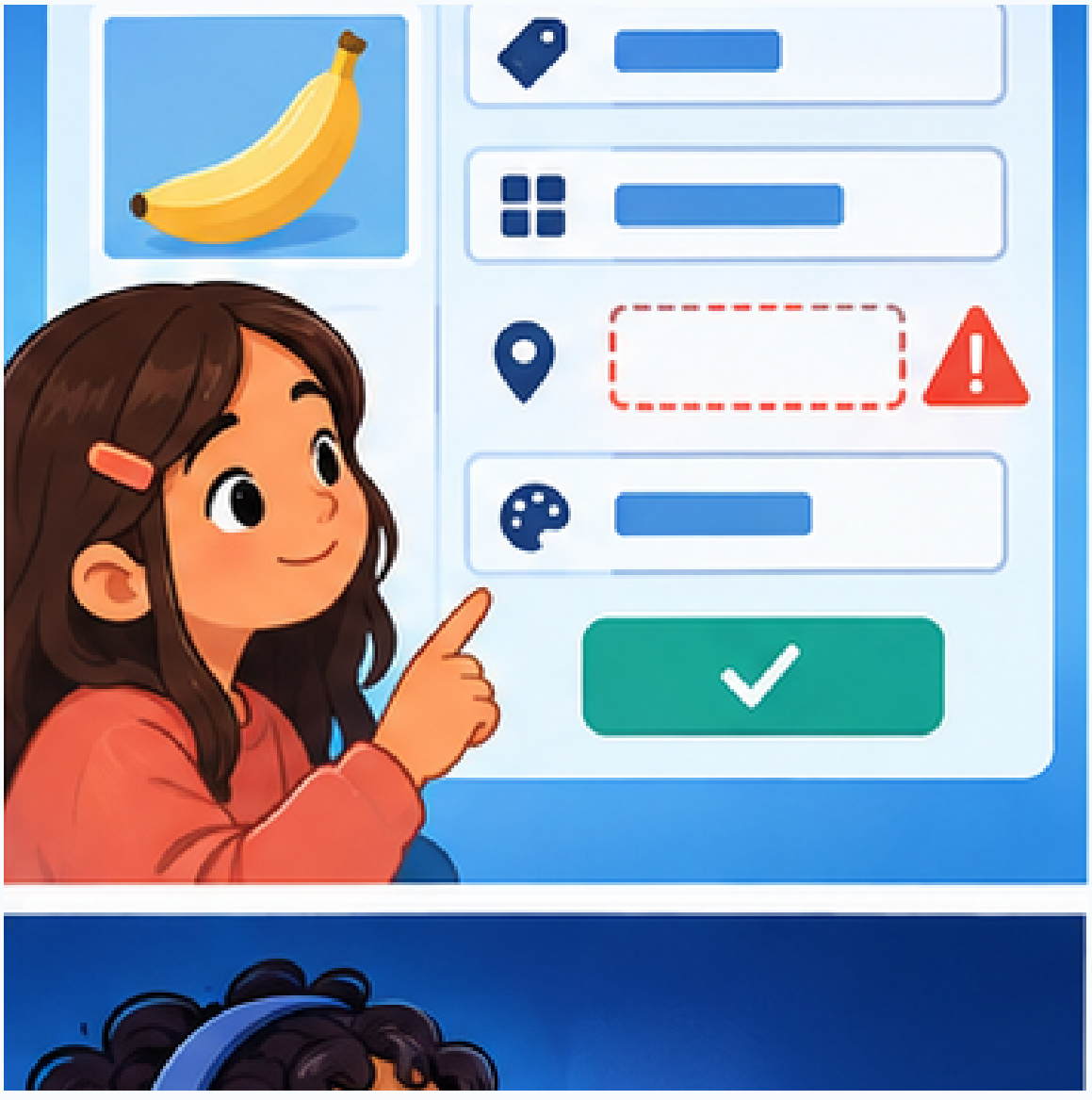
Without opening the old lesson, explain one key idea from Prompt as a Contract. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from A Distribution of Possibilities. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 8: Structured **Output** and **Schema**s

Schemas make **output** easier to validate, reuse, and test.



CREATOR MISSION

Request **field**s, then check missing and invalid values.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

> Does a requested **output** format prove a response followed the specification?

This week's end goal

By the end, I can validate structured **output** in layers: syntax, **schema** and source meaning.

STUDENT LESSON · 2 OF 18

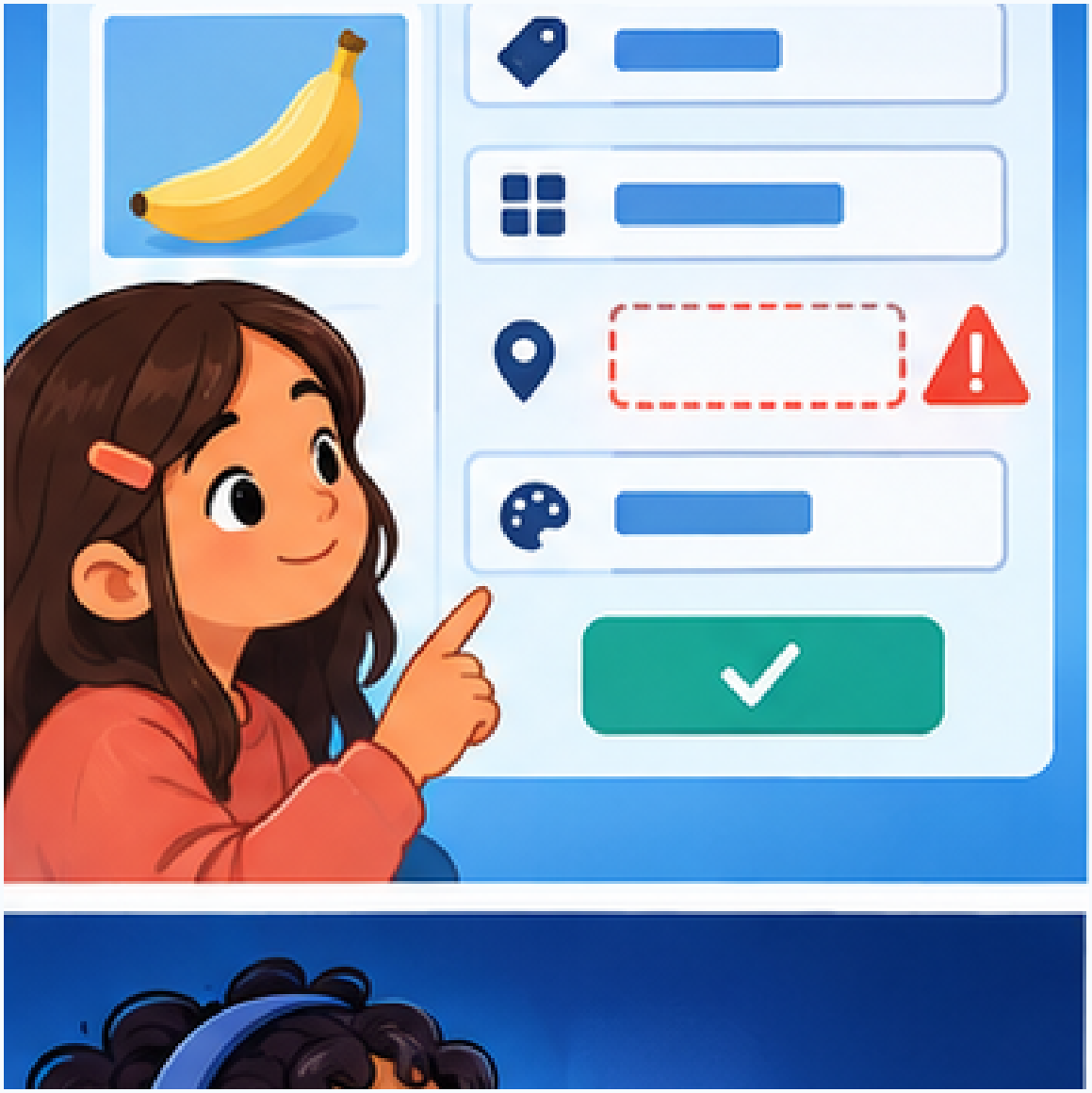
Notice the evidence

> Investigate the supplied Structured Output and Schemas case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use structured **output** and **schema**s to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

A response may be valid JSON and still violate the **schema** or sources. First parse the text. Then check the exact object **field**s and **type**s. Finally check source support and domain meaning.

These checks answer different questions. Do not call a successful parser a complete validator.

Use the Week 11 closed object: exactly status,room,capacity,source_ids. status is supported or needs_review. source_ids is a list of distinct known source-ID strings.

For supported, room is a nonempty string and capacity a nonnegative integer. For needs_review, room and capacity are null. Boolean values are not capacities under this contract.

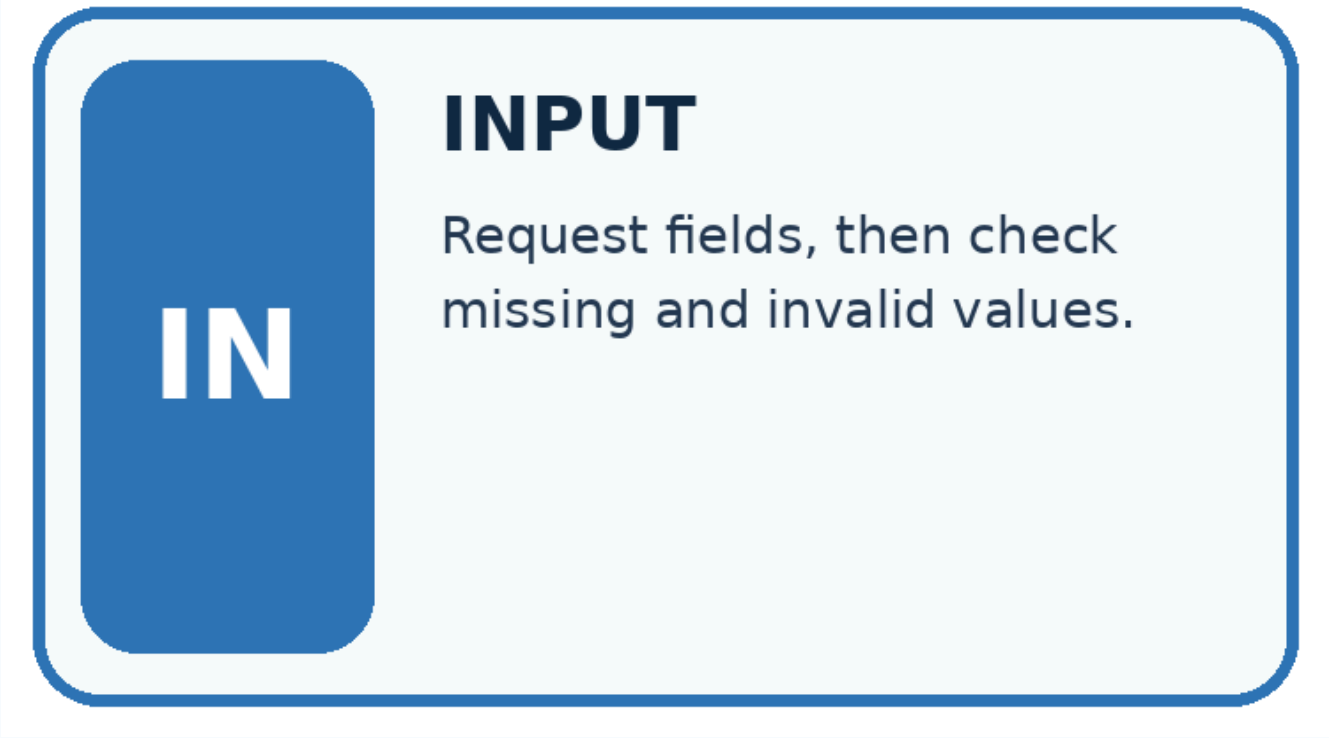
See how it works

> Follow the mechanism for Structured Output and Schemas. Connect each labelled input, deciding step and result.

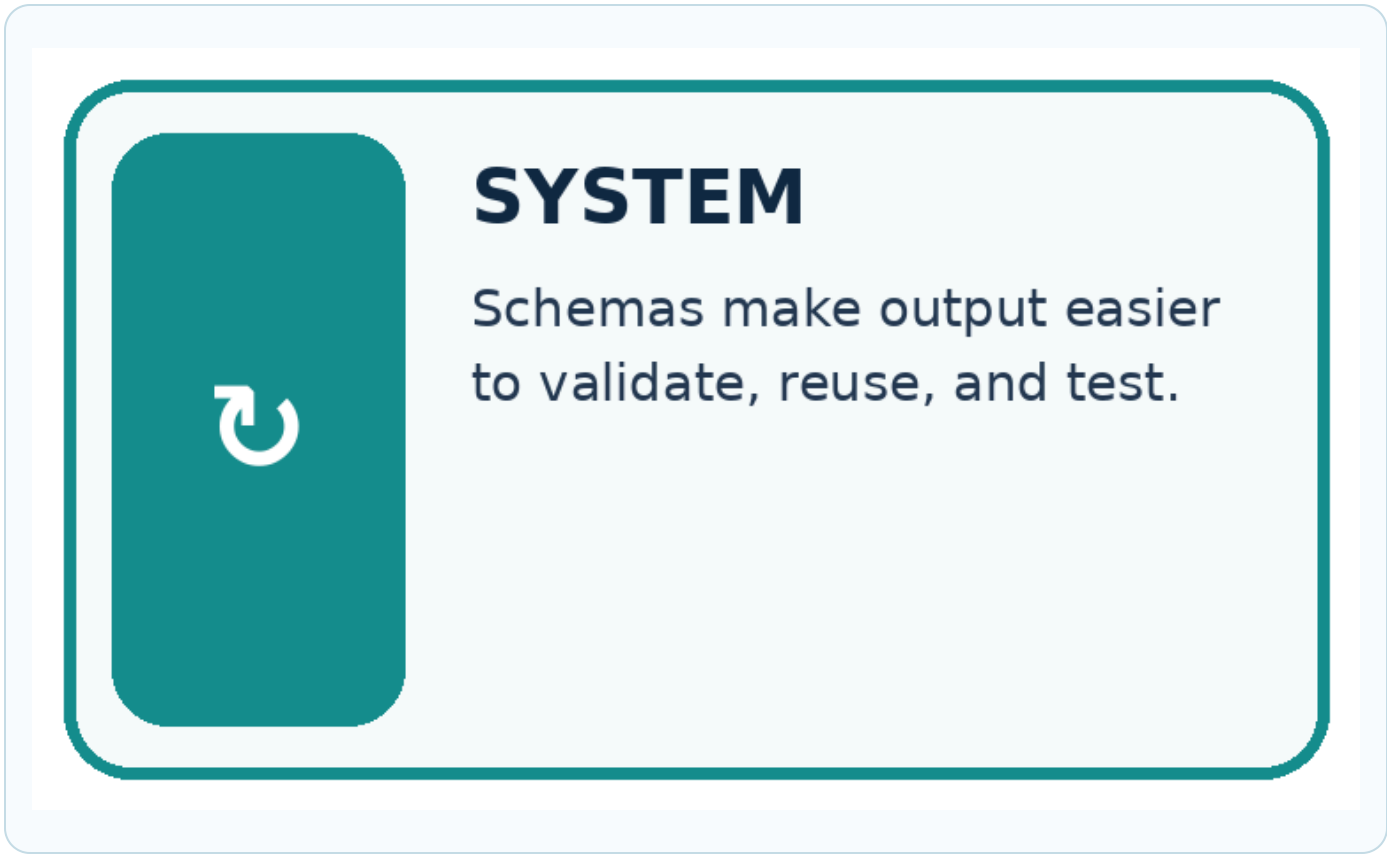
VISUAL MECHANISM

Input → Process → Output

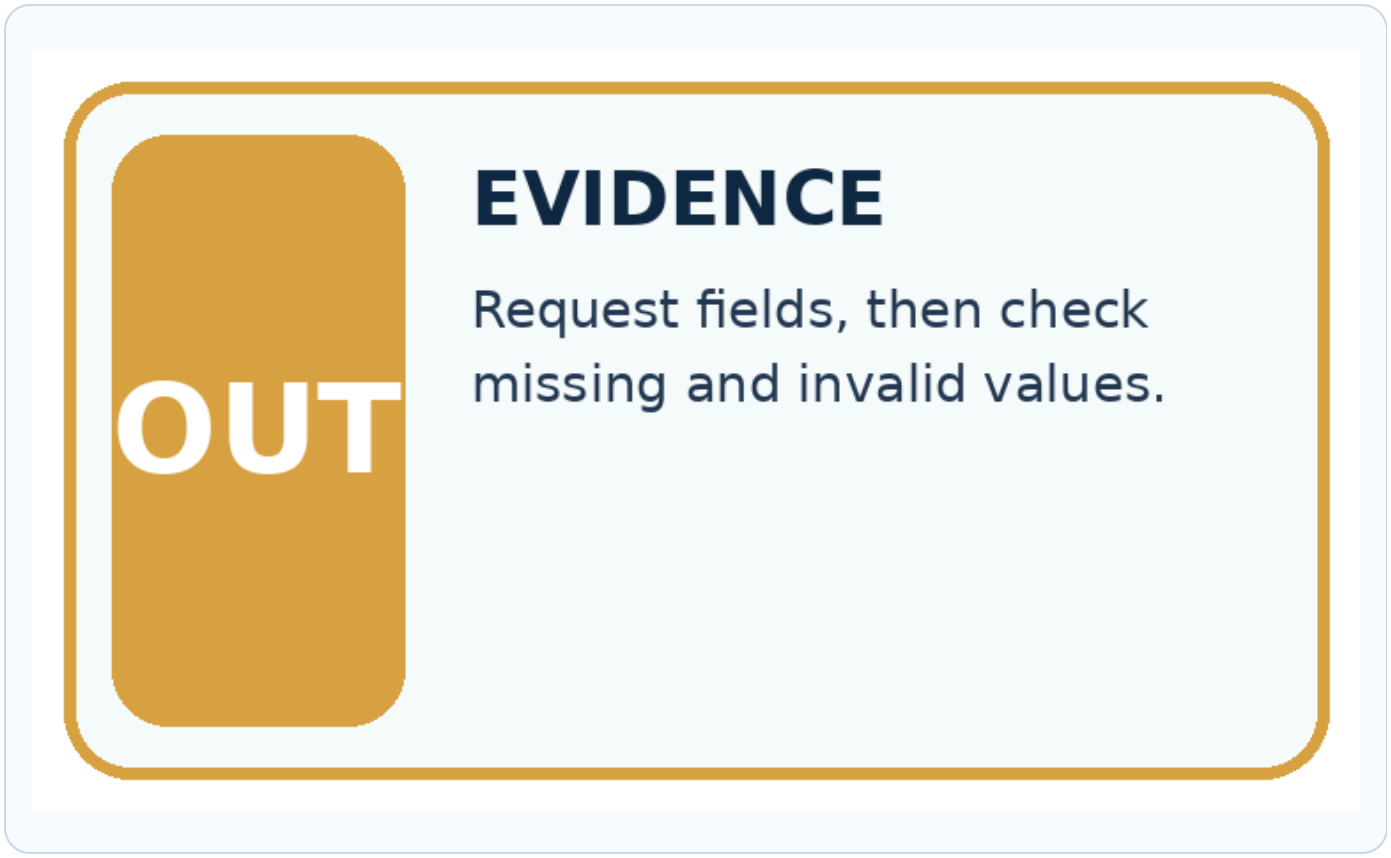
Trace the information. At each stage, ask what changed and how you could test it.



↓


SYSTEM
Schemas make output easier to validate, reuse, and test.

↓


OUT
EVIDENCE
Request fields, then check missing and invalid values.

Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

Response fragment	Layer that rejects it	Reason
{status: supported}	Syntax	JSON keys/strings need quotes
{"capacity":"12"}	Schema in a complete object	String does not satisfy integer capacity
{"capacity":true}	Schema in a complete object	Boolean is not accepted as an integer
An extra booking field	Closed schema	Unexpected field

These fragments illustrate failures, not complete candidate objects. A real check receives the whole object and verifies every required **field**.

Do not repair a string capacity automatically unless a separate documented conversion policy permits it; otherwise report **validation** failure. A null value differs from missing a **field**.

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Generated text	JSON parse	A value or syntax error
Parsed value	Closed-schema check	Contract match or schema error
Contract-valid object	Match claims to current sources	Supported or semantic error

Each passed layer still leaves later layers to check.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

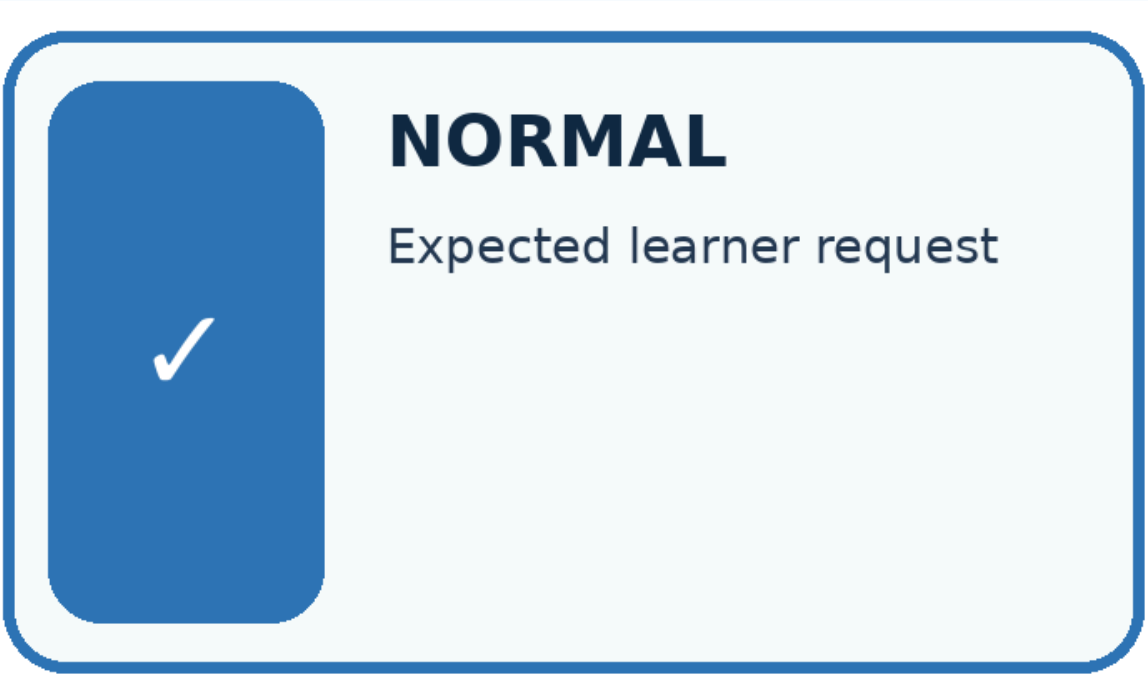
> Compare the supplied cases for Structured Output and Schemas. Identify the detail that changes the conclusion.

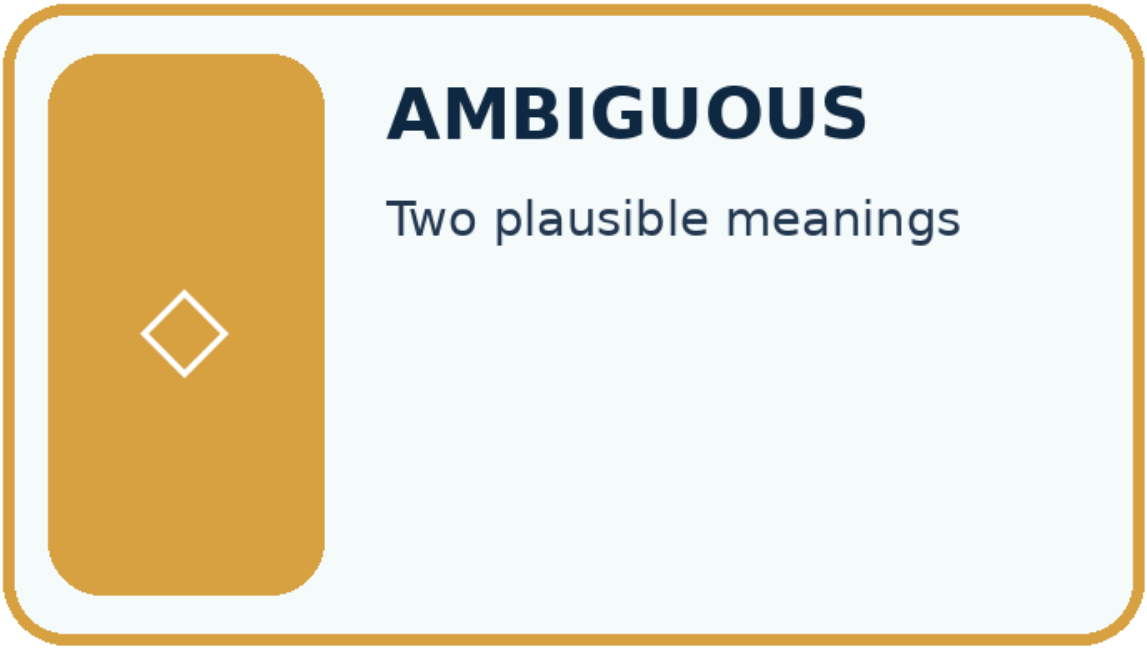
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Add an extra **field** "rating":5 to X under the exactly-**field**s **schema**.

Predicted result

Prediction: The changed object fails **schema** despite keeping the correct page count.

Why the result follows

Reason: Exactly means additional **field**s are not allowed in this contract.

STUDENT LESSON · 5 OF 18

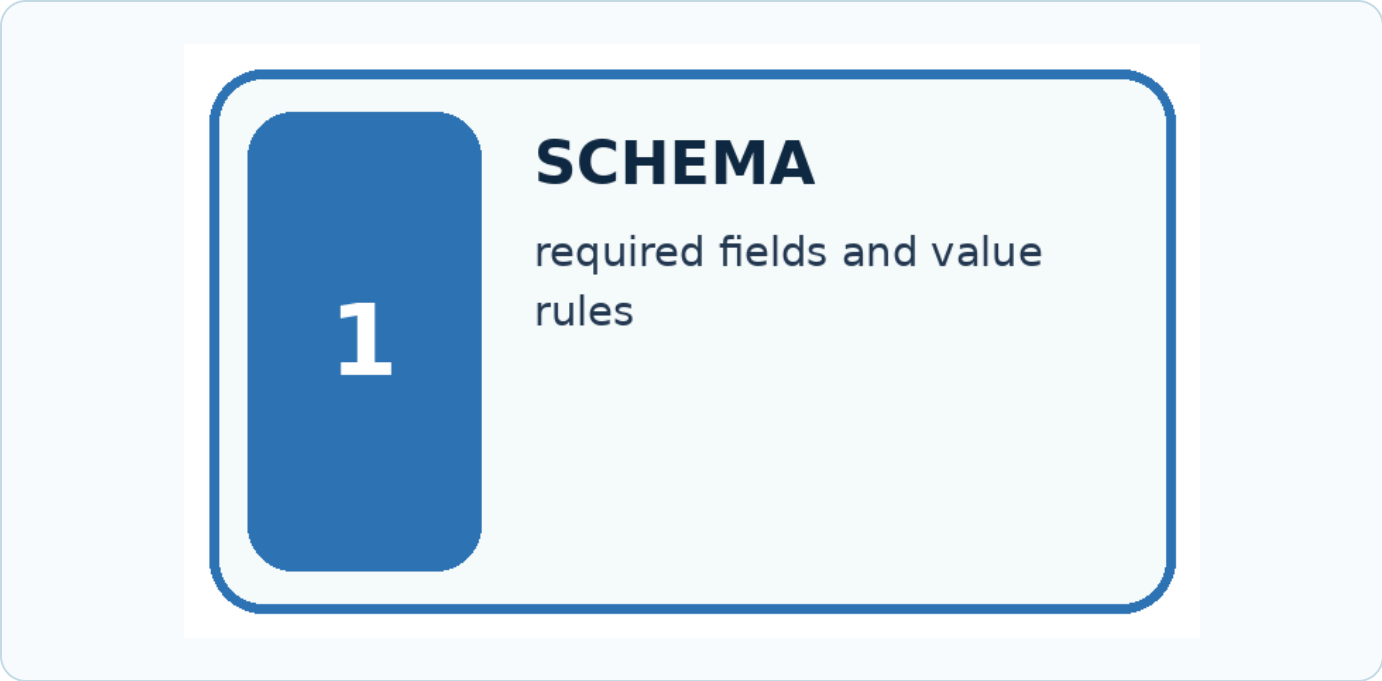
Words that unlock the idea

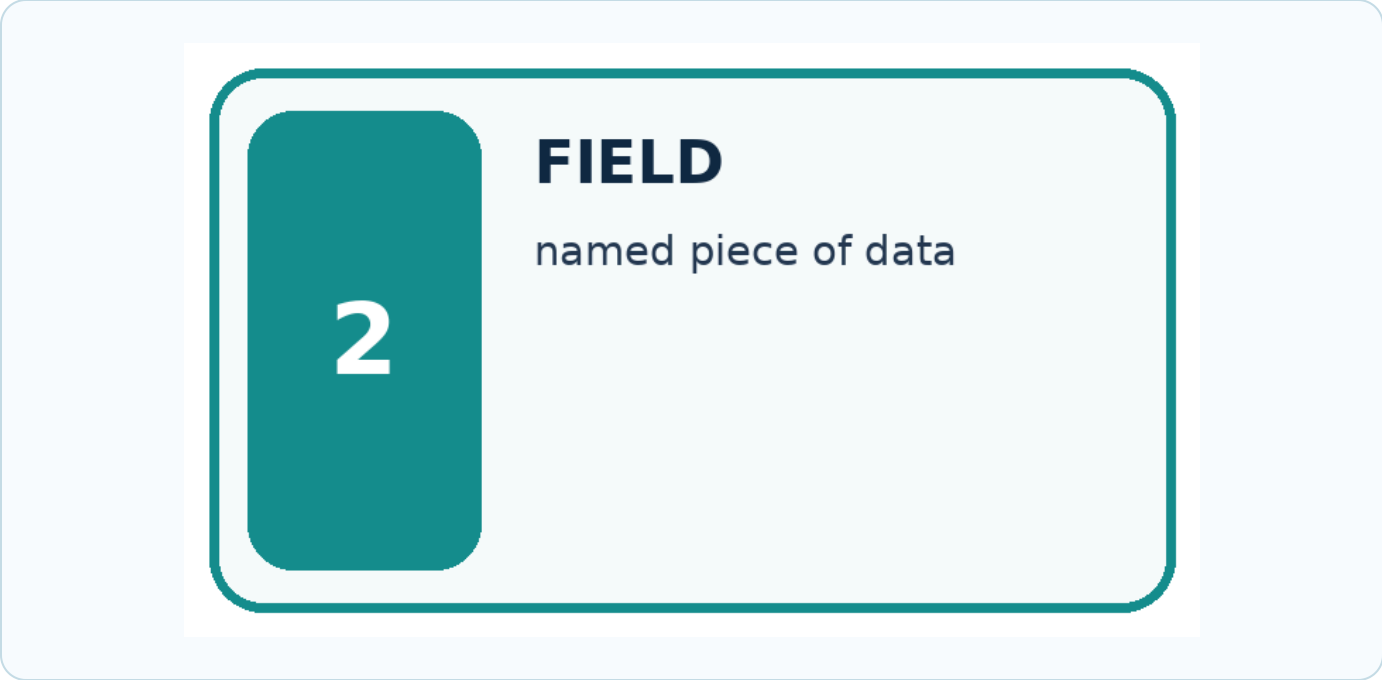
> Use the vocabulary of Structured Output and Schemas to explain a concrete decision, rather than only memorising definitions.

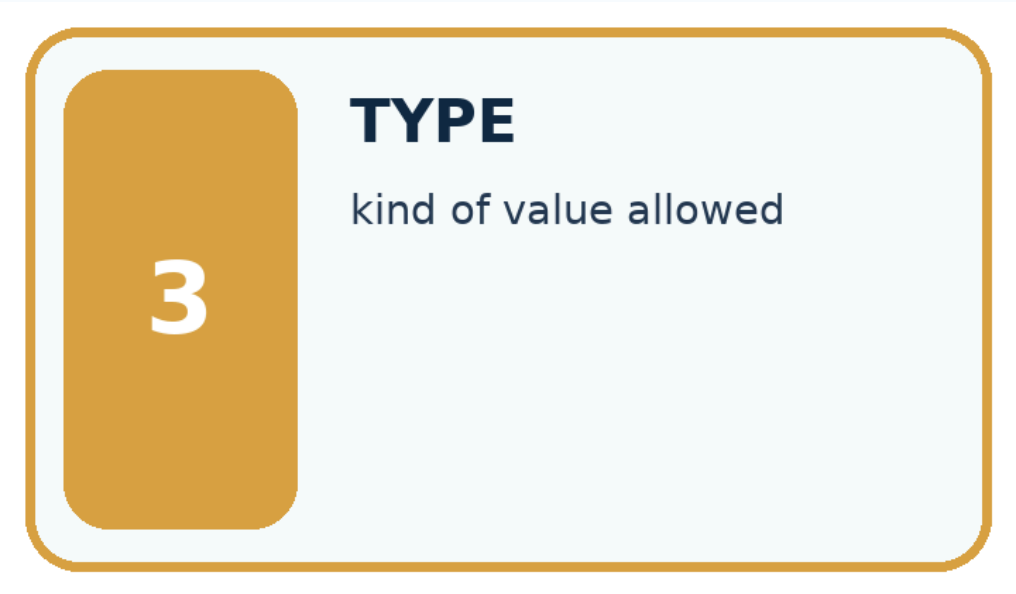
VOCABULARY LAB

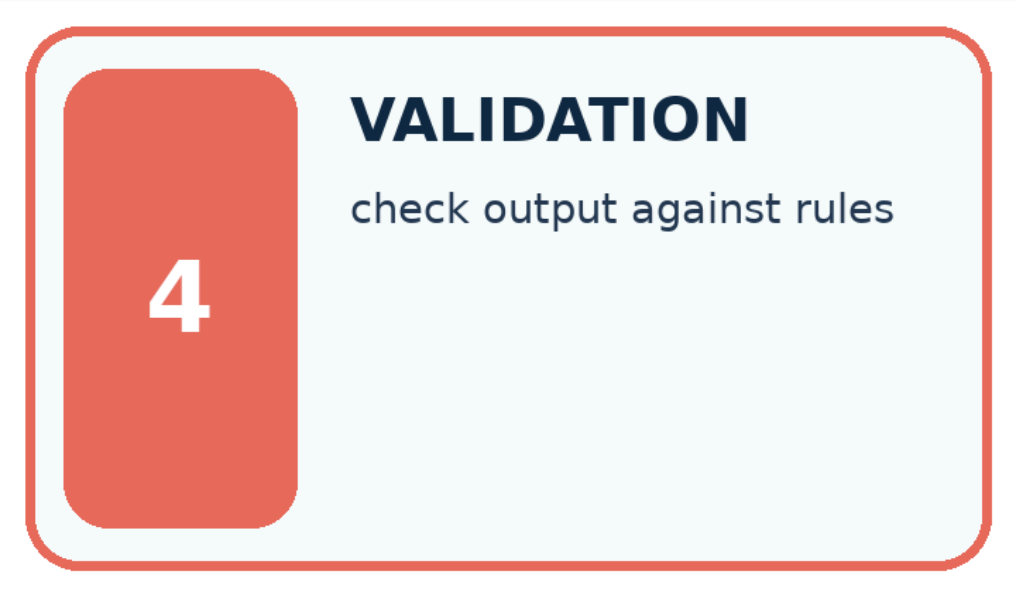
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
SCHEMA
required fields and value rules


2
FIELD
named piece of data





Word	My own example	Not this / boundary
schema	A specification of required fields, types and structural rules for data.	
field	A named item of information in a record or structured message.	
type	The category of a value, such as text, number or Boolean.	

Word	My own example	Not this / boundary
validation	Checking data or behaviour against rules; in ML, validation data is used to choose settings or models, separately from final testing.	

Words used in the closer explanation

Word	Meaning
Syntax validation	Checking whether text can be parsed under the format grammar
Schema validation	Checking fields, types and allowed combinations in a parsed value
Semantic validation	Checking meaning and consistency with sources or domain rules
Closed schema	A contract rejecting fields beyond the stated set

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STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Structured Output and Schemas, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 8 • WORKED EVIDENCE

1. Schema fields: answer=text; evidence=list; confidence=low/medium/high.
2. Output omits evidence: validation fails.
3. Confidence=97 is the wrong type and allowed-value set.
4. Reject, repair, or request again before downstream use.

1. **Schema field**s: answer=text; **evidence**=list; confidence=low/medium/high.

2. **Output** omits **evidence**: **validation** fails.

3. Confidence=97 is the wrong **type** and allowed-value set.

4. Reject, repair, or request again before downstream use.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Worked explanation · Structured Output and Schemas

Suppose sources P and Q approve room B and give capacity 12.

The complete object {"status":"supported","room":"B","capacity":99,"source_ids":["P","Q"]} parses and satisfies the stated **field type**s, but its capacity is unsupported by Q. Reject it at **semantic validation**.

Listing a real source ID does not prove that source supports the associated claim.

Worked explanation · Structured Output and Schemas

An object with needs_review and room B violates the conditional **schema** rule even if its source IDs are real. An object with supported, room B and capacity 12 but source ID Z fails the known-source rule.

Require the linked current approval and capacity records to support the same room.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Structured Output and Schemas

A fictional Study Buddy returns a machine-readable book fact.

The facts we will use

Supplied facts

Schema: exactly title text, pages integer, source_id text. Approved record R: River Notes, 64 pages.

Supplied facts

X={"title":"River Notes","pages":64,"source_id":"R"}; Y uses pages:"64"; Z uses pages:46 with otherwise identical **field**s.

AI ACADEMY / AI-3/8

Structured Output and Schemas

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Parse

Each shown object is valid JSON syntax.



2. Schema check

Y fails because its pages value is text, not an integer.



3. Meaning check

Z fits field types but contradicts R's page count.



4. Accept X

X passes syntax, declared schema and source comparison.

CHANGE ONE THING

Add an extra field "rating":5 to X under the exactly-fields schema.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Parse	Each shown object is valid JSON syntax.
2. Schema check	Y fails because its pages value is text, not an integer.
3. Meaning check	Z fits field types but contradicts R's page count.
4. Accept X	X passes syntax, declared schema and source comparison.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Parsing establishes that text forms a structured object. **Schema** checks establish required **field**s and **type**s. **Semantic validation** compares their meaning with the approved source.

These are different layers: a parseable, well-shaped object can still contain a false fact. Report the first failing layer without treating format success as factual success.

What this example does not establish

Keep the stated limit

Source agreement does not establish that the approved catalogue itself is correct or complete.

Keep the stated limit

Read the labelled picture: Which object demonstrates that valid JSON and valid **type**s are insufficient for truth?

Keep the stated limit

Change one thing: Add an extra **field** "rating":5 to X under the exactly-**field**s **schema**. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Structured Output and Schemas claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: Valid-looking braces prove the content is correct. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Keep **validation** outcomes and the original generated text for fictional debugging data.

Report the first failing layer and enough detail to locate the issue; later **validation** should not be reported as passed when it was never reached. Even a fully valid reporting object does not authorise a booking.

The application retains the read-only permission boundary.

Exit: a parser accepts JSON, but capacity is true and an extra book_now field is present. State which checks remain and why no booking should occur.

STUDENT LESSON · 8 OF 18

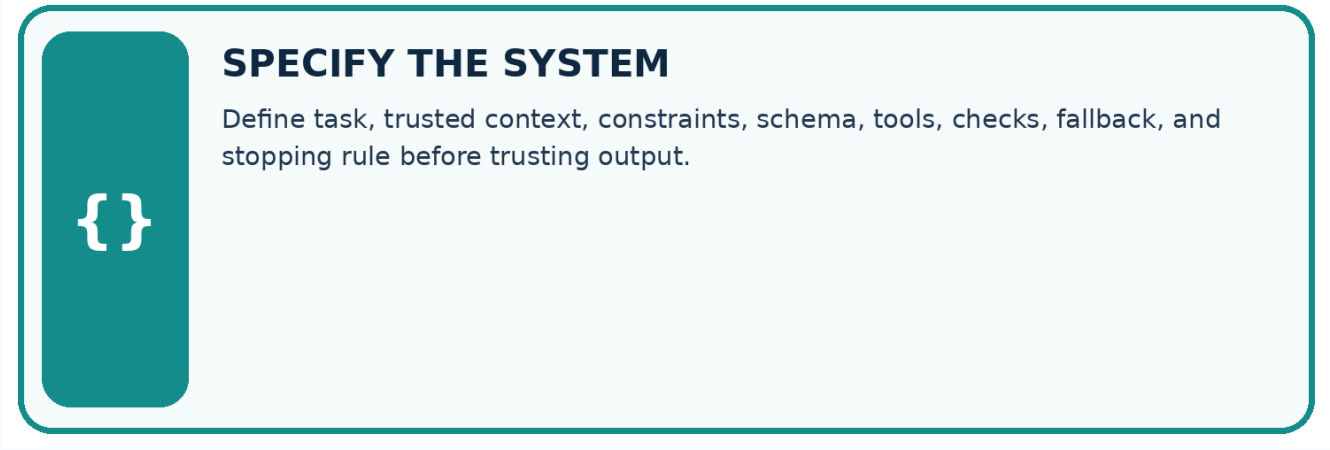
Practise together

> Practise Structured Output and Schemas with support, then explain the deciding step in your own words.

LOGIC STUDIO

Specify Before You Run

Prompts, **schema**s, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output_schema** LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE **field**s, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Supported object {"status":"supported","room":"D","capacity":8,"source_ids":["P","Q"]} matches approval P for D and register Q for D capacity 8. It passes all three layers.

Replacing capacity with 0 still passes **schema** because zero is nonnegative, but fails source consistency unless Q actually reports zero. Preserve measured zero rather than treating it as missing.

Guided: name the layers passed and explain why a schema-valid zero still needs source support.

Practise with less help: Audit types and fields

Use workbook W2 for the connected example “Structured **output**s and **validation**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Structured **output**s and **validation**. Proposed explanation: “Valid JSON means correct answer”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which object demonstrates that valid JSON and valid **type**s are insufficient for truth?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

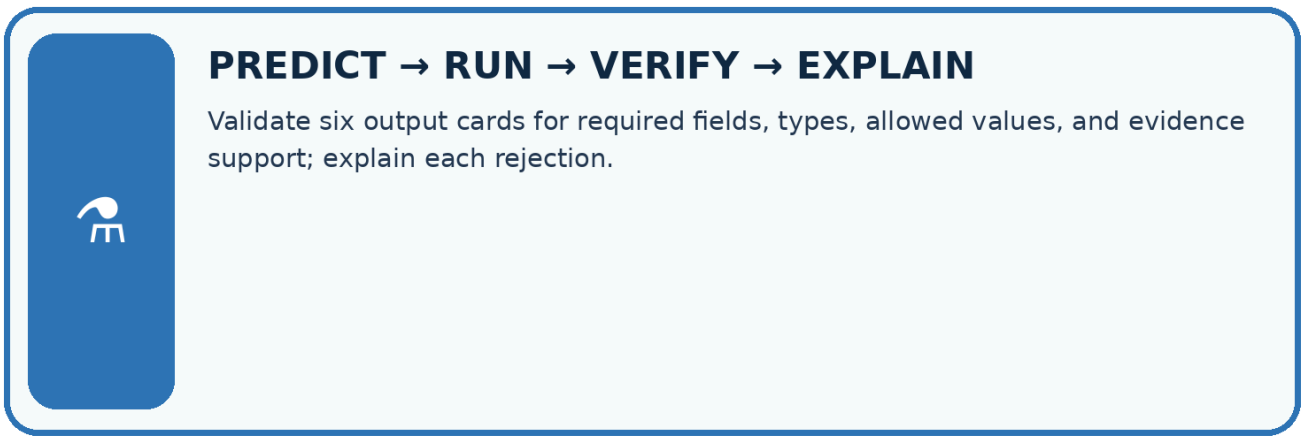
Run the investigation

> Investigate Structured Output and Schemas. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Validate six **output** cards for required **field**s, **type**s, allowed values, and **evidence** support; explain each rejection.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

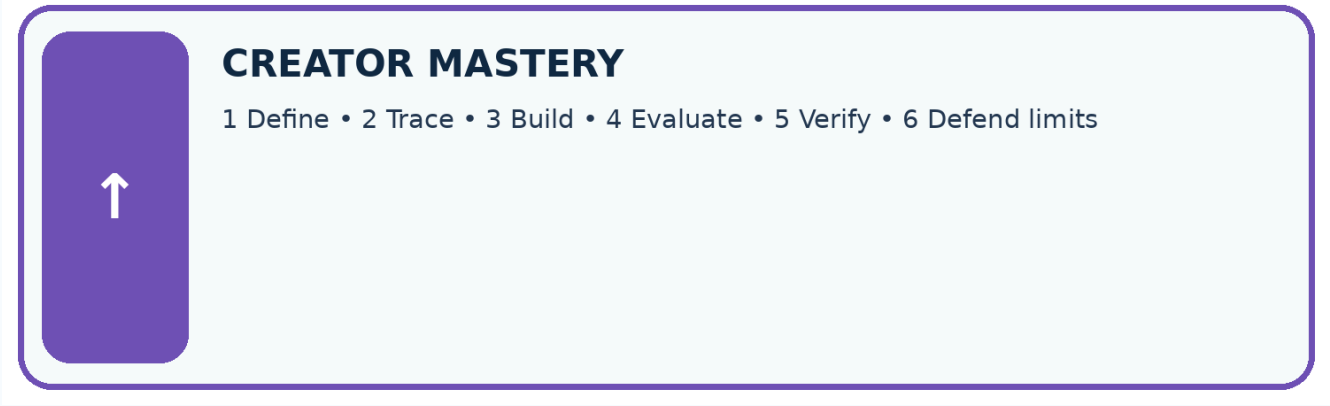
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Structured Output and Schemas. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

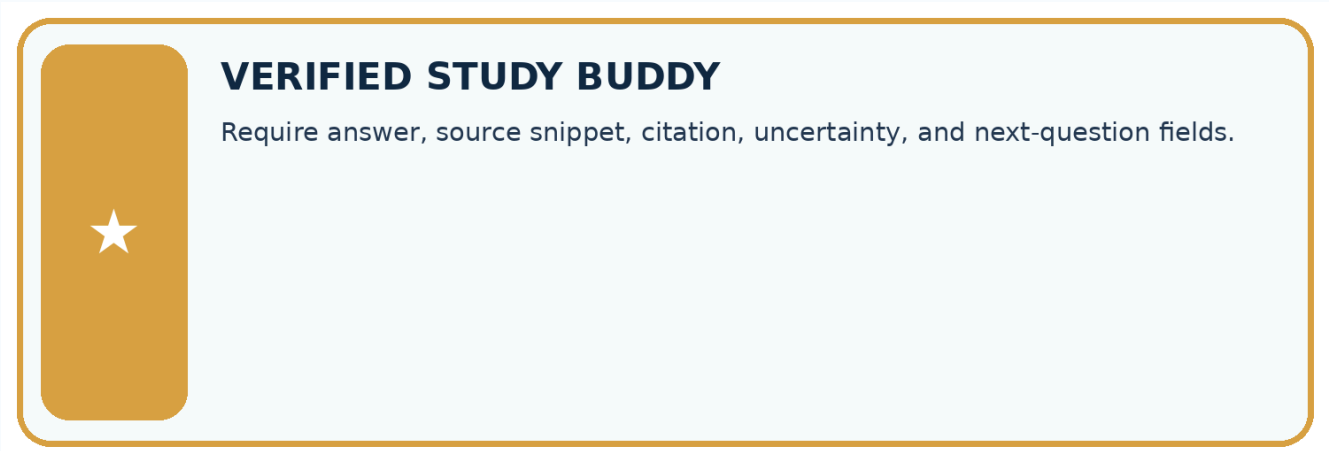
Connect your project

> Apply this week's idea to Verified Study Buddy. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Require answer, source snippet, citation, uncertainty, and next-question **field**s.

Trace the system

SYSTEM MAP Learner need → prompt/**schema** → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

Evidence for your project

Validate Study Buddy **output**s before another program consumes them; retain source IDs and separate syntax/**type**/support failures.

STUDENT LESSON · 12 OF 18

Show what you know

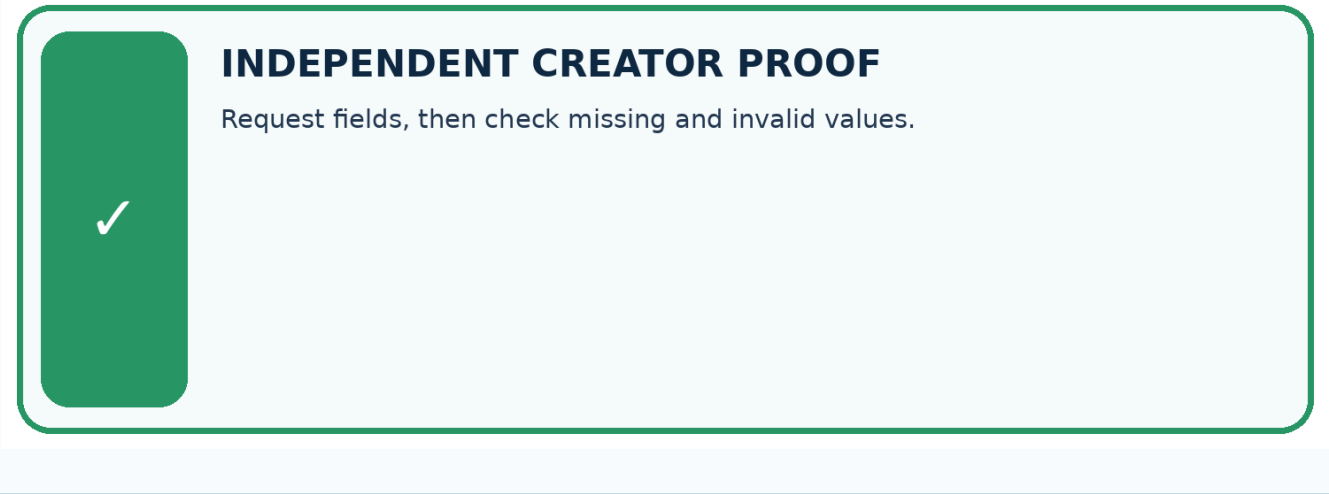
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Request **field**s, then check missing and invalid values.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Structured Output and Schemas case for supported practice. Repeated examples do not become fresh assessments.

Week 8 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Schema requires title text and pages number. **Output** is {"title":"River","pages":"80"}.

The question

Diagnose the shape error.

Worked explanation

Pages is a string, not a number. Use numeric 80. Structure **validation** does not verify the real page count.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

Trace the supplied case visually

Type correctness is not factual correctness

Evidence or stage	Value or result
Wrong type	pages = "80" is text
Correct type	pages = 80 is numeric
Remaining check	Verify the real page count

Schema requires title text and pages number. **Output** is {"title":"River","pages":"80"}.

Diagnose the shape error.

Pages is a string, not a number. Use numeric 80. Structure **validation** does not verify the real page count.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Validate new objects independently

Sources A approve room F and B register F capacity 9. Object X has supported,F,9,[A,B] with exactly the required **field**s. Y replaces 9 with "9". Z keeps integer 9 but names room G.

Trace the first failing layer for each and explain the difference.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 8 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A required **schema** has title as text and count as an integer. The **output** is {"title":"Seeds","count":"three"}. Identify the error and give a valid replacement.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Require answer, source snippet, citation, uncertainty, and next-question **field**s.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

> **Schema** requires title:text and pages:number. **Output** is {"title":"Cloud","pages":"80"}. Fix its structural error. Would the repaired object prove the book really has 80 pages?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Structured Output and Schemas, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional Study Buddy returns a machine-readable book fact.

By the end, I can	How I will show it
Separate JSON parsing, schema validation and source support checks	Explain the deciding step, show a trace or test, and state one limit.
Reject invalid types and unexpected fields under a closed schema	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Trace a validly formatted but unsupported claim	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Structured Output and Schemas without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Separate JSON parsing, **schema validation** and source support checks

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Study Buddy returns a machine-readable book fact.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? X passes syntax, declared **schema** and source comparison.

1. Meaning check

2. Accept X

3. **Schema** check

4. Parse

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Accept X X passes syntax, declared **schema** and source comparison. Parsing establishes that text forms a structured object.

Schema checks establish required **field**s and **type**s. **Semantic validation** compares their meaning with the approved source. These are different layers: a parseable, well-shaped object can still contain a false fact.

Report the first failing layer without treating format success as factual success.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Add an extra **field** "rating":5 to X under the exactly-**field**s **schema**. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. The changed object fails **schema** despite keeping the correct page count.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The changed object fails **schema** despite keeping the correct page count. Exactly means additional **field**s are not allowed in this contract.

Repair a missing building block

Review Iterative Prompting (ai-3/7). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: What **evidence** supports improvement here, and what remains unproven?.

Return to this week and explain what you changed.

Review Prompt as a Contract (ai-3/5). Goal: Write a prompt specification with observable response criteria. Short practice: Why should the tool not fill melting from memory even if it knows a common definition?.

Return to this week and explain what you changed.

Review A Distribution of Possibilities (ai-3/2). Goal: Check whether a next-token table is a valid probability distribution. Short practice: Why is 0.625 not a broken probability sum?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

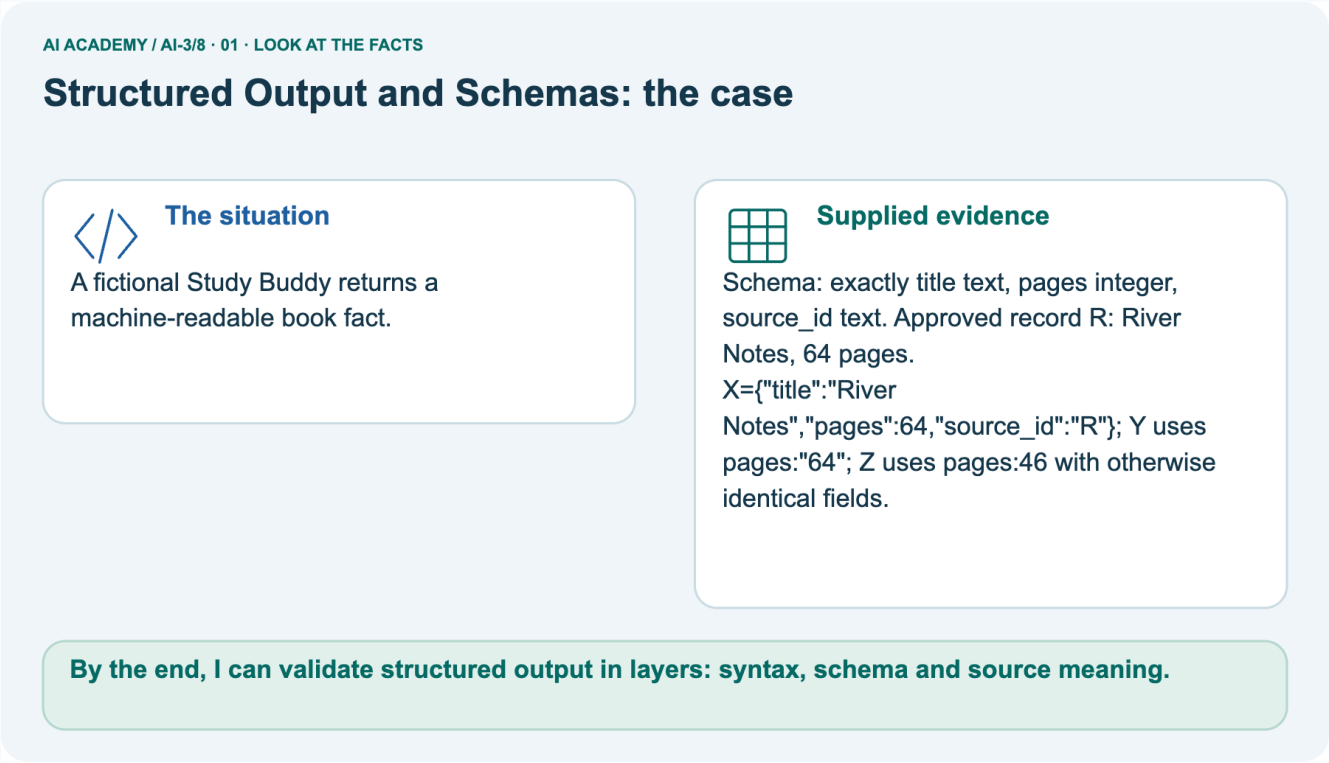
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Structured Output and Schemas graphic using its labels, deciding evidence and relevant limitation.

01. Structured Output and Schemas: the case



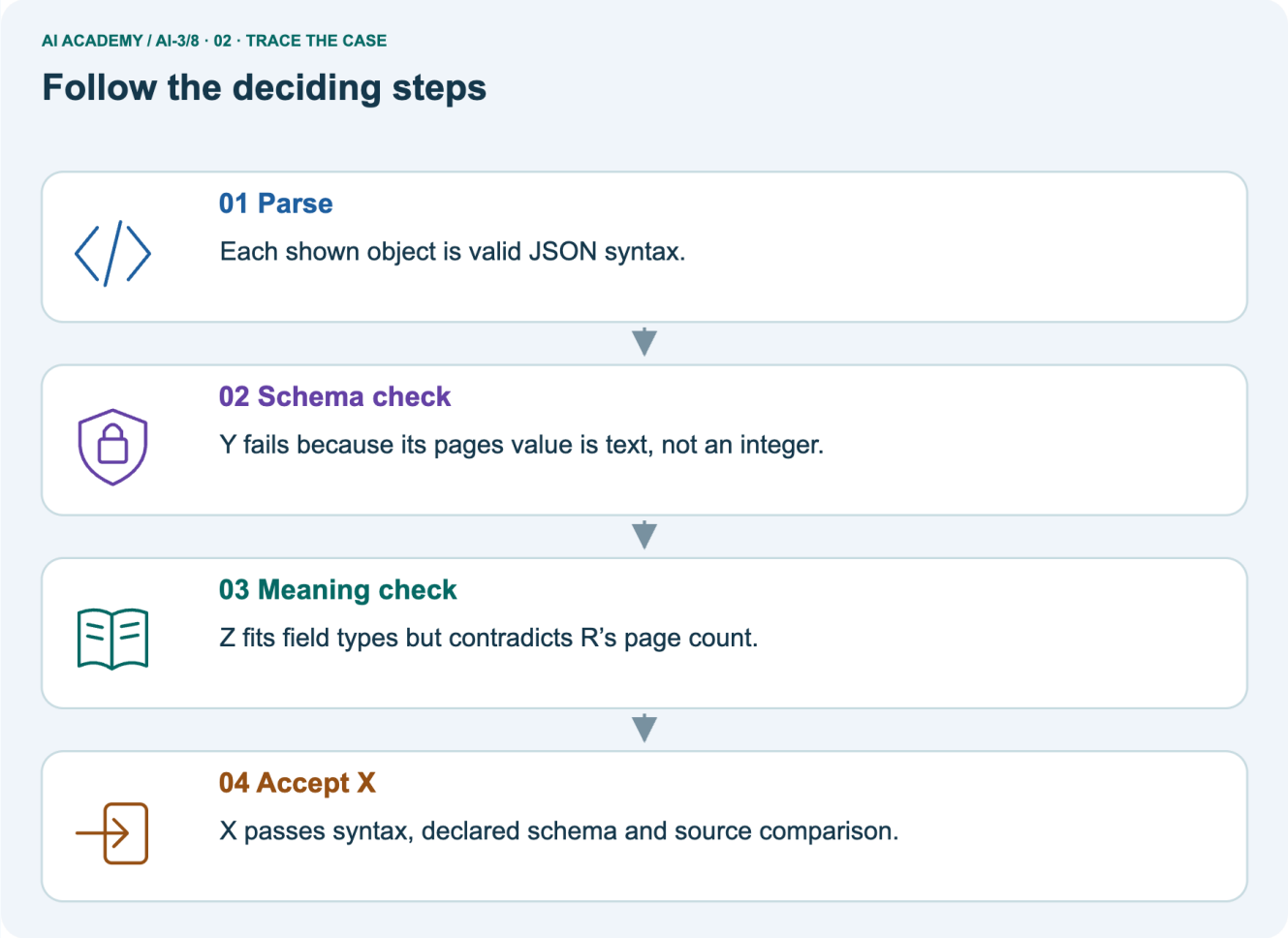
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional Study Buddy returns a machine-readable book fact. Supplied facts: **Schema**: exactly title text, pages integer, source_id text. Approved record R: River Notes, 64 pages.

X={"title":"River Notes","pages":64,"source_id":"R"}; Y uses pages:"64"; Z uses pages:46 with otherwise identical **field**s.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

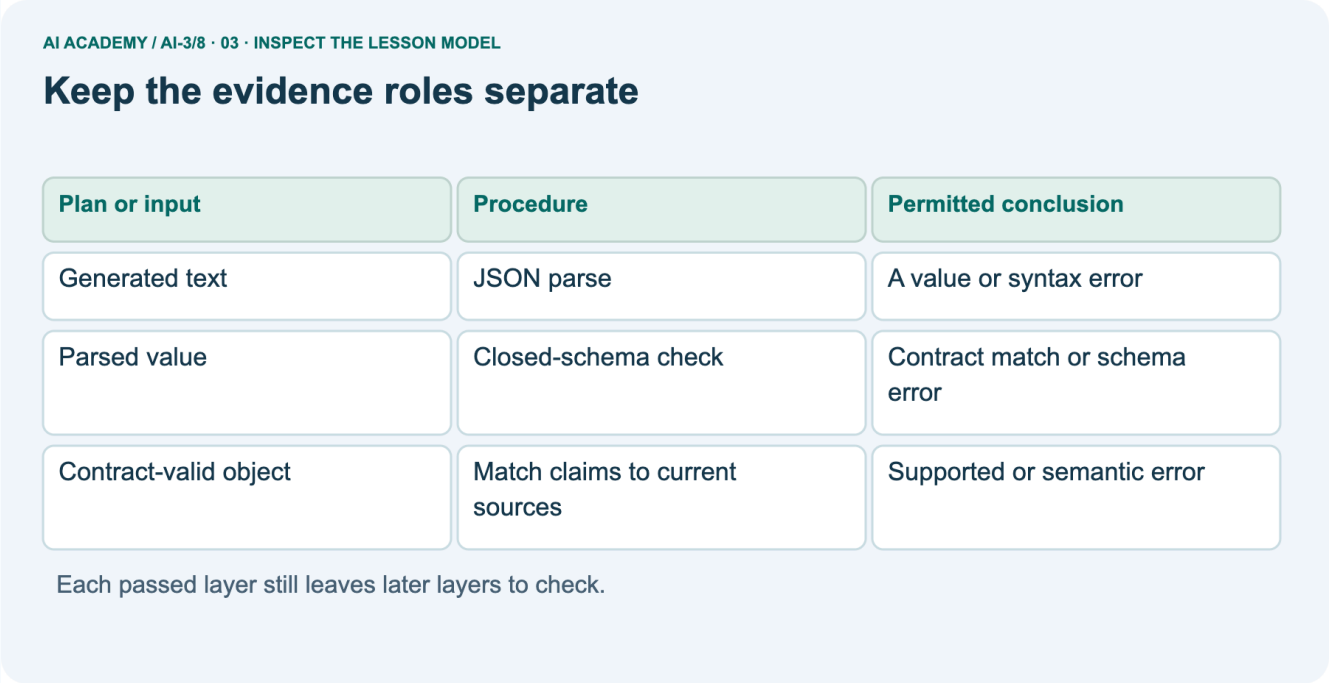


Follow the labelled picture with the matching explanation. Open the vector diagram

Parse: Each shown object is valid JSON syntax. → **Schema** check: Y fails because its pages value is text, not an integer. → Meaning check: Z fits **field type**s but contradicts R's page count. → Accept X: X passes syntax, declared **schema** and source comparison.

Point to each numbered stage and explain which recorded **input** it uses.

03. Keep the evidence roles separate



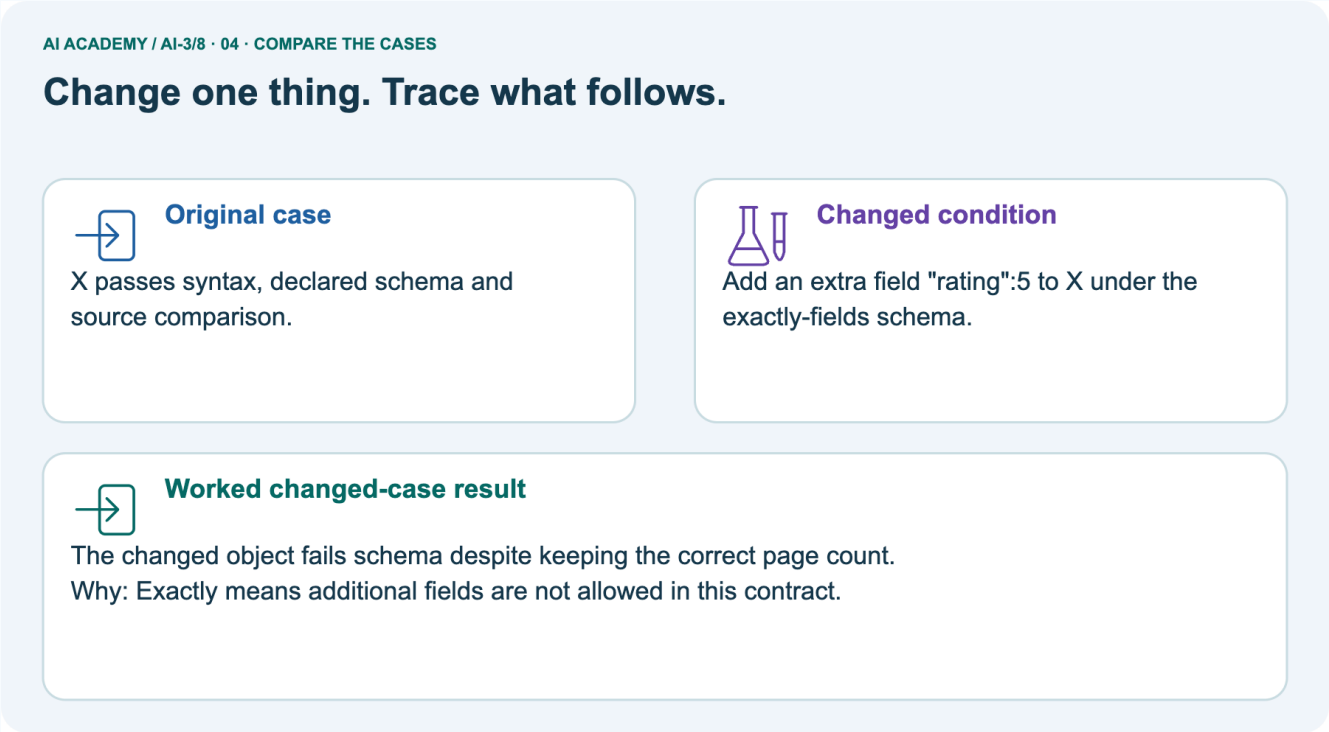
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the **evidence** roles separate. Generated text | JSON parse | A value or syntax error. Parsed value | Closed-**schema** check | Contract match or **schema** error.

Contract-valid object | Match claims to current sources | Supported or semantic error

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



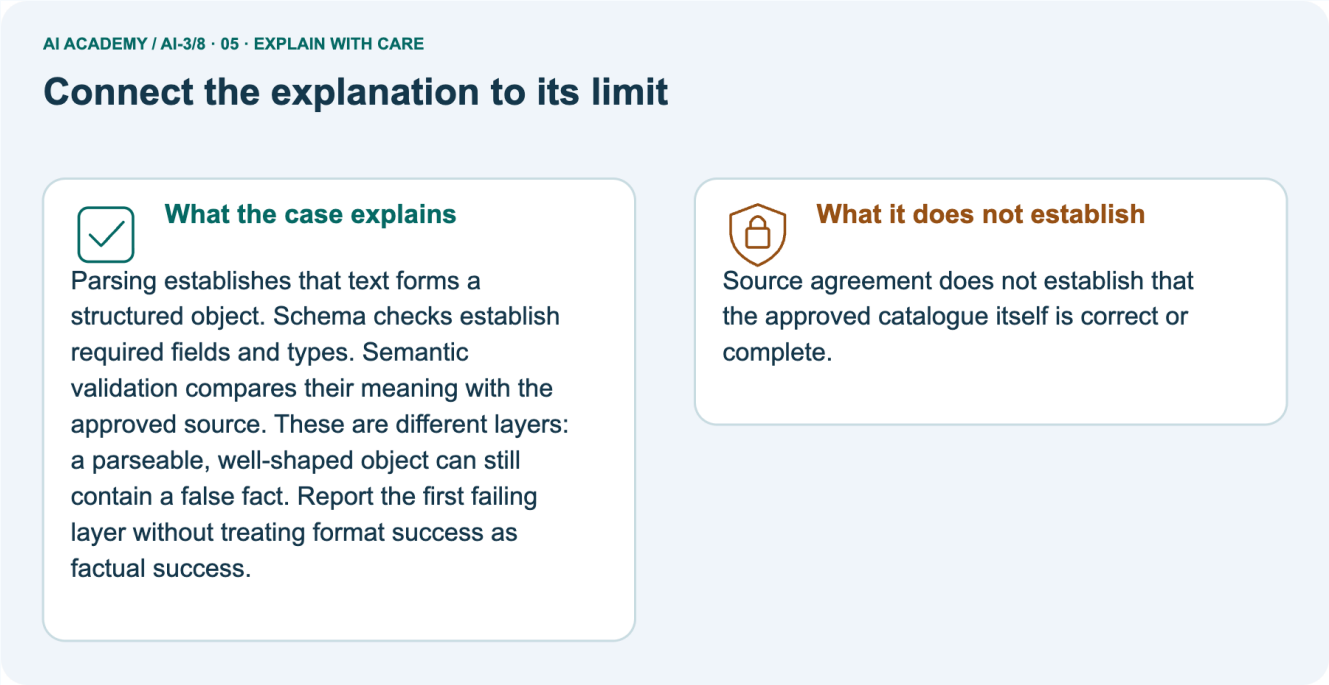
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: X passes syntax, declared **schema** and source comparison. Changed condition: Add an extra **field** "rating":5 to X under the exactly-**field**s **schema**.

Expected result: The changed object fails **schema** despite keeping the correct page count. Reason: Exactly means additional **field**s are not allowed in this contract.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

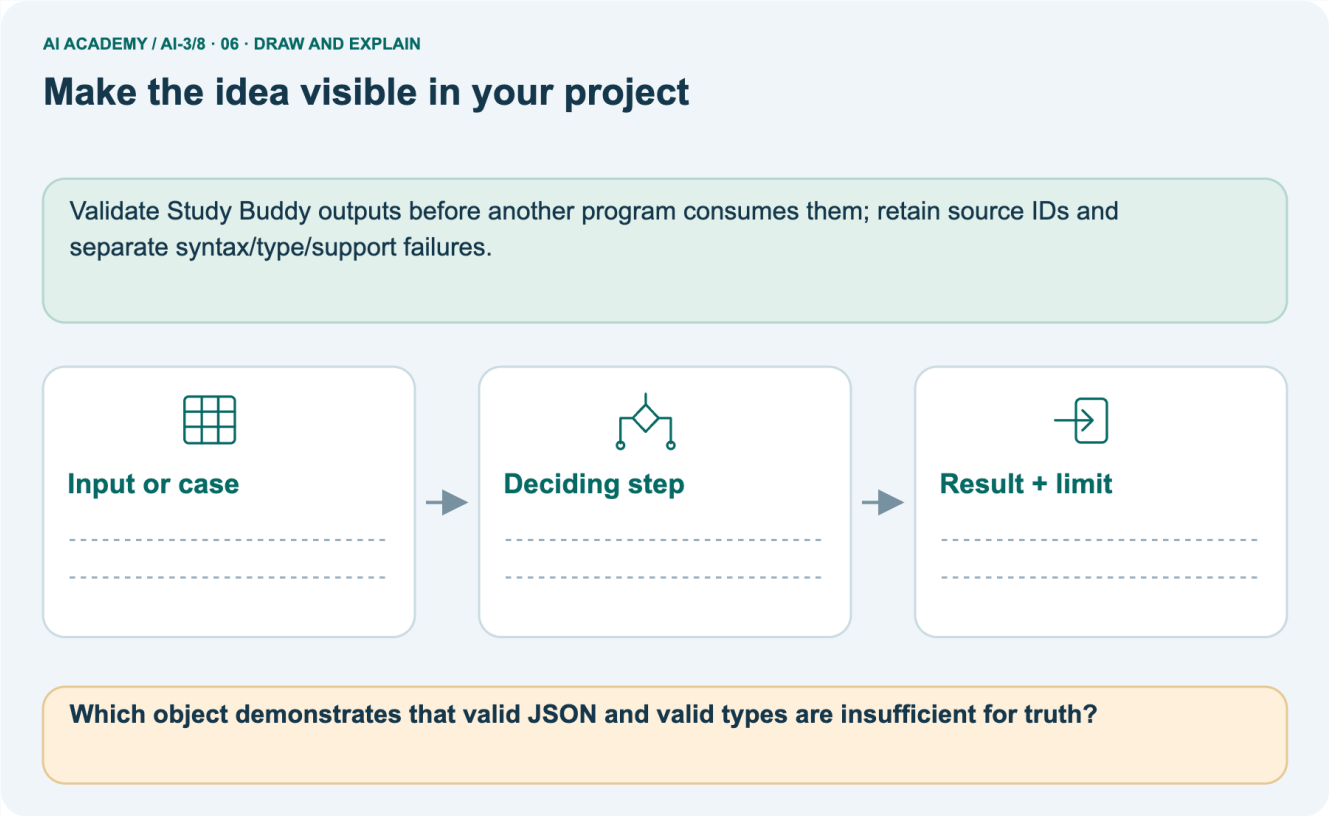
Parsing establishes that text forms a structured object. **Schema** checks establish required **field**s and **type**s. **Semantic validation** compares their meaning with the approved source.

These are different layers: a parseable, well-shaped object can still contain a false fact. Report the first failing layer without treating format success as factual success.

Limit: Source agreement does not establish that the approved catalogue itself is correct or complete.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Validate Study Buddy **output**s before another program consumes them; retain source IDs and separate syntax/**type**/support failures. Draw your **input**, deciding step, result and limit.

Supported question: Which object demonstrates that valid JSON and valid **type**s are insufficient for truth?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.